

A discrete iteration related to Pierce expansions

OPG-819

L1 - TRACTABLE

PARTIALLY SOLVED

Number Theory

ORIGINAL PROBLEM STATEMENT

Conjecture Let $a > b > 0$ be integers. Set $b_1 = b$ and $b_{i+1} = a \bmod b_i$ for $i \geq 0$. Eventually we have $b_{n+1} = 0$; put $P(a, b) = n$.

Example: $P(35, 22) = 7$, since $b_1 = 22, b_2 = 13, b_3 = 9, b_4 = 8, b_5 = 3, b_6 = 2, b_7 = 1, b_8 = 0$.

Prove or disprove: $P(a, b) = O((\log a)^2)$.

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